

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a)      | method for measuring a curved claw (on the image) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 |
| 2(b)(i)   | <p><i>any <b>one</b> from:</i></p> <p>colour of disc ;</p> <p>material of disc ;</p> <p>type of particle ;</p>                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 |
| 2(b)(ii)  | <p>(level of) roughness</p> <p><b>or</b></p> <p>(mean) particle diameter (on disc) ;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 |
| 2(b)(iii) | 24 <b>or</b> 25 (feeding visits) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 |
| 2(b)(iv)  | <p><i>any <b>two</b> from:</i></p> <p>1 as the particle diameter increases, (percentage of feeding visits when buff-tailed bumblebees) gripped (on to the disc) increases ; <b>ora</b></p> <p>2 at 30 <math>\mu\text{m}</math> and 53 <math>\mu\text{m}</math> / over 30 <math>\mu\text{m}</math>, there is no, difference / change / increase / AW (in percentage of feeding visits when buff-tailed bumblebees gripped on to the disc) ;</p> <p>3 3. at 5 <math>\mu\text{m}</math>, (buff-tailed bumblebees) only hover ;</p> | 2 |

| Question  | Answer                                                     |                                            |                                                                                     |                                                                                             |                         |       | Marks |
|-----------|------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------|-------|-------|
| 2(c)(i)   | Table 2.2 completed                                        |                                            |                                                                                     |                                                                                             |                         |       | 2     |
|           | mean particle diameter on the disc / $\mu\text{m}$         | rank of mean particle diameter on the disc | percentage of feeding visits when the buff-tailed bumblebees gripped on to the disc | rank of percentage of feeding visits when the buff-tailed bumblebees gripped on to the disc | Difference in rank, $D$ | $D^2$ |       |
|           | 5                                                          | 1                                          | 0                                                                                   | 1                                                                                           | 0                       | 0     |       |
|           | 9                                                          | 2                                          | 4                                                                                   | 2                                                                                           | 0                       | 0     |       |
|           | 12                                                         | 3                                          | 16                                                                                  | 3                                                                                           | 0                       | 0     |       |
|           | 16                                                         | 4                                          | 79                                                                                  | 4                                                                                           | 0                       | 0     |       |
|           | 30                                                         | 5                                          | 99                                                                                  | 5.5                                                                                         | 0.5                     | 0.25  |       |
|           | 53                                                         | 6                                          | 99                                                                                  | 5.5 ;                                                                                       | 0.5                     | 0.25  |       |
|           |                                                            |                                            |                                                                                     |                                                                                             | $\Sigma D^2 = 0.5 ;$    |       |       |
| 2(c)(ii)  | $(r_s =) 0.99 / 0.986 ;$                                   |                                            |                                                                                     |                                                                                             |                         |       | 1     |
| 2(c)(iii) | 0.99 / calculated value for $r_s$ , is greater than 0.83 ; |                                            |                                                                                     |                                                                                             |                         |       | 1     |